

Building a Learning Organization in the Offshore Oil Industry

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The meek may inherit the earth, but not its mineral rights.
Source: J.P. Getty

The oil industry has always drawn a distinction between the intangible assets that are often the source of underlying value in oil exploration and production, and mere physical assets. In more recent years, it has, for example, become clearer that successful exploration for oil or gas reserves is related to the quality and private nature of the information that a corporation acquires about a possible hydrocarbon reservoir, and the manner in which that information is manipulated and interpreted. A corporation's capabilities in information and knowledge acquisition and management are, therefore, critical distinguishing features in long term competition for new oil and gas reserves.

In the 1990s, there has been a new emphasis on the way that corporations can learn to identify, appraise and develop reservoirs—each an essentially unique geological configuration—in the most effective manner. This paper reports on lessons concerning strategic learning capabilities encountered in exploratory case studies in UK Continental Shelf oil field development projects. The case studies particularly highlight four aspects of strategic learning capabilities: a strategic concern with institutionalizing an appropriate learning style; an emphasis on the pace of learning as a competitive tool; the structuring of learning capabilities so as to recognize

Since the collapse of the price of crude oil in the mid 1980s, some participants in the sector have overhauled their approach to learning for strategic advantage. The management teams dealing with the projects discussed here demonstrated an awareness that learning can be a core rigidity as well as a core competence, and set about institutionalizing *new* learning. The speed of learning was an important determinant of success, leading to substantial increases in project NPVs and a rapid reinvestment in the learning capabilities themselves. Learning about 'boundary management' helps to distinguish areas where internal learning could develop proprietary competitive advantage, from those areas where valuable learning might best be achieved through alliance. © 1999 Elsevier Science Ltd. All rights reserved

areas where the potential for competitive advantage exists; and the structuring of learning to recognize other areas where collaborative approaches to learning through alliances provide better returns.*

*The underlying case studies were conducted as a research for a joint Open University Business School/BBC business video, entitled "A New Way of Life". This video has been broadcast and is available as part of the OUBS MBA course, "Strategy" (OU, 1997, Milton Keynes). It also won a silver medal at the New York Film Festival.

Background to the Case Studies

The studies were limited to the upstream exploration and production sectors of the oil and gas business. These sectors are not concerned with the



transportation, trading or refining of crude oil or gas, nor with the marketing and retailing of oil, gasoline or petrochemical products.

The study reported on here was based primarily on a series of interviews with senior management of three offshore oil fields in the UK Continental Shelf. In all cases the operator of the fields was BP, although interviews were also conducted with BP's partners, allies, and suppliers. The fields are distinguished from each other by the stage of 'upstream' activity underway at the time of the study (see Fig. 1). Taken together, the three fields provide examples of most pre-production activities in the upstream oil sector.

Upstream oil and gas businesses primarily comprise three types of activities:

- Crude oil or natural gas supplies must be discovered. This exploration activity was represented in the study by the Schehalion field, together with the recent history of discovery of the Foinaven field. These two fields were discovered after 1992, and are likely to prove to be the first commercial oil fields to be exploited in the hostile and environmentally sensitive Atlantic 'frontier' west of the Shetland Islands.

- Once a reservoir has been discovered, oil companies must ensure that its reserves are capable of being commercially exploited. This spans both the appraisal of the recoverable quantity of oil or gas in the reservoir, and the process of developing the most effective production infrastructure. The Foinaven field was in the process of simultaneous appraisal and development, while the third field (Andrew) was nearing the completion of its development. Andrew was originally discovered in the mid-1970s, but remained unexploited until a new alliance-based approach to its development rendered it commercially viable in the 1990s.
- The third upstream activity—physical extraction of oil and gas, or 'production'—was not an activity that featured centrally in the study.

Upstream oil and gas is characterized by the key role played by information and knowledge in exploration and appraisal, and by high levels of uncertainty throughout. Uncertainty arises mainly from political, geological, geophysical and engineering unknowns concerning the physical reservoir, the best means to maximize the oil recovery rate, and the financial uncertainties surrounding the long-term realizable revenue stream. Much upstream activity traditionally occurs in partnerships or joint ventures, utilizing a wide range of suppliers and contractors. Information and knowledge, where not carefully guarded, can spread quickly through the sector.

Institutionalizing Appropriate Learning

Business has seen an explosion of interest in the area of organizational learning. Senge¹ has identified the new task of leaders as the creation of the conditions for organizational learning, while Pedler et al.² have suggested methods to encourage organizations to become 'learning companies'. Others see the need to challenge existing mental models or shared cognitive structures as a key activity for management; de Geus,³ from his experiences of using scenarios at Shell, argues that this is the key purpose of effective strategic planning.

Learning as a competitive weapon was most famously explored by the Boston Consulting Group's development of the experience curve.⁴ Interest in the strategic value of learning has blossomed, however, alongside increased interest in the resource-based explanation of competitive success,⁵ which views learning as a complex, potentially distinctive strategic capability. Organizational learning, by its nature, arises from a complex interaction of individual and team learning skills. It is, therefore,

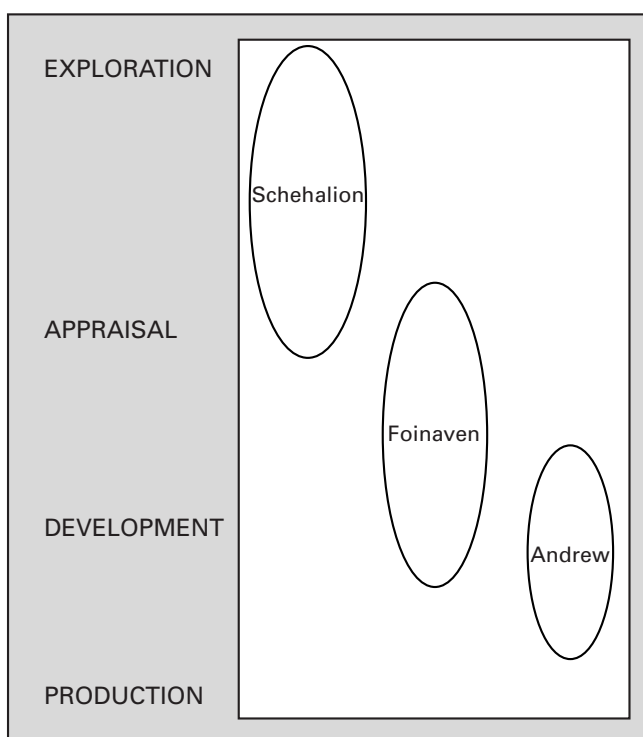


FIGURE 1. Status of the oilfields in the mid-1990s.

TABLE 1. Traditional approach to learning

CHARACTERISTICS	OUTCOMES
● Importance of pre-project reports and studies ● Perceived need to address all uncertainties ● Preservation of many options ● Heavily professionalized ● Separate inputs from functional departments ● Significance of post-project audit	● Marginal reduction in uncertainty ● Value of options often illusory ● High appraisal and development costs ● Encouraged a sequential approach to development ● Prolonged appraisal and development phases ● Reduced the present value of projects

less likely to be capable of being imitated, and more likely to prove an enduring source of competitive advantage. Prahalad and Hamel,⁶ for example, argue that the core competencies that underpin global corporate competition reflect 'collective learning' of how to integrate and co-ordinate complex skill sets with sophisticated technologies.

These writers reflect the increase in corporate learning activities such as external benchmarking, reverse engineering, total quality management, continuous improvement programmes, intranets and corporate knowledge systems. Mission statements increasingly refer to the objective of becoming a 'learning organization'.

This interest in organizational learning, and its identification as a potential source of competitive advantage, may, however, encourage the view that all learning is advantageous. Challenging this, Leonard-Barton⁷ describes how knowledge sets in the area of innovative new product development can become 'core rigidities', rather than core competencies.

In the cases studied in this report, learning has not been discovered as a new corporate capability. Oil companies have always valued knowledge and learning in upstream exploration and production. The traditional approach to learning has, however, been one that has emphasized the production of professional reports by the companies' own engineers, geologists, surveyors, and geophysicists. Considerable time was spent producing detailed pre-project reports, designed to identify and assess most areas of project uncertainty, and enumerate and evaluate multiple development options. This approach of learning everything learnable about the potential development led to long delays between discovery and the production of first oil, reinforced by the sequential approach taken to studying the uncertainties of development. The professional quality of these reports was usually undeniable, the products of an institutionalized approach to high quality learning which is summarized in Table 1.

The reported outcomes were, however, less desirable.

This approach was being replaced in the companies studied, however, by one that recognized the limitations of uncertainty reduction. Instead, the new approach emphasized the need to identify those key factors, which ensure the success or failure of a project. Possible outcomes for such factors were then identified and contingency strategies were designed to be put in place should the project be threatened. This was matched by an emphasis on action learning as a method of assessing uncertainties.

Breaking free of the reliance on detailed professional pre-project assessment has led to three further changes in the approach to learning:

- The sequential approach to risk assessment and development planning has been replaced by an approach, which allows parallel exploration, appraisal and development. The only impediments to parallel proceedings are created where key project success factors must be assessed. This parallel approach has significantly shortened the discovery-to-first-oil cycle time for fields such as Foinaven and Schehalion (see Fig. 2).
- Information, knowledge and learning have been able to flow more rapidly, through the creation of task-orientated teams and lateral linkages between professionals.
- The systematic studies which emphasized the pre-project views of the companies' own specialist professionals, have given way to an approach which accesses the best available skills as and when they are needed, opening up alliancing as a key strategy to enhance learning. This new approach to learning is summarized in Table 2.

The difference between the two approaches to learning is also captured in Box 1, an example quotation from BP's Asset Manager (the chief executive of the Foinaven and Schehalion fields)

Box 1**The little cloud of problems**

"Let's consider that every new discovery is a little cloud of problems. You may think that discovering an oil field is a moment of celebration, the work is finished, and BP has made another excellent discovery. How it shows up for the teams of people working on the discovery, however, is it is a little cloud of problems. How are we going to solve these things? How are we going to take this discovery and move it through to the point where we produce barrels of oil. Now there are two ways to deal with this little pile of problems called the discovery.

Let's take Schehalion: Schehalion lies in very deep water; in a harsh climate; its an interesting reservoir but with fairly heavy oil; there are huge environmental concerns. All these things need to be dealt with. Originally what we would have done is pool together quite a large team of technologists, geologists, engineers, and so forth, and work every problem to death. Now by working every problem to death, I mean writing learned reports on every problem, be they small or big problems—putting them together in beautifully prepared reports, getting people like me, the manager, to sign off, saying, 'this is a very important piece of work'.

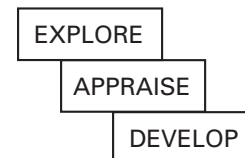
Inevitably those things end up in files or libraries, and sometimes some person comes along with a lot of money and makes an investment in this cloud of problems, for all it is still a cloud of problems for which people have written learned reports. The investment decision is taken, the project is developed, the field goes on production and it doesn't behave the way people expect it to behave. We then get the post-project audit report—the famous post-project audit—in which people write down why it didn't work, and that in turn is filed, and we never learn. Libraries are full of this type of learning.

There is an alternative though. Let's take the same problems, the same little cloud of concern that the Schehalion discovery is. Lets look into that cloud with teams who are trained to look at what's going to make a difference here. Not everything is going to make a difference. There will be some problems at Schehalion which will be there now, and they'll be there during production, but they're not going to make a great deal of difference to the profitability of the project. So we distinguish those problems that are going to make a difference if intervened against: which are going to make the project highly profitable, and which problems, if we do nothing about, are going to kill us. We get a very small sub-set of that big cloud of concern, which used to be the discovery—a little subset of those problems that are going to make a difference. Then we go to work on these problems—not understanding them for their own sake, not writing learned reports about them, but actually inventing interventions to put in place should those problems come about."

Colin MacLean, BP Asset Manager.

(a) Traditional Learning: Sequential Activities

9 to 10 years

(b) New learning: Parallel Activities

4years

FIGURE 2. Reducing the discovery-to-first-oil.

was critical if the inappropriate, traditional approach was not to remain a core rigidity.

Formal structures and systems were changed to allow information and learning to flow more freely across the organization. Corporate structures were refocused on business units based on individual oil fields or groups of oil fields, termed simply 'Assets'. Geologists, for example, reported on the benefits of understanding reservoir engineers needs gained by simply working at desks alongside them, instead of within functional departments. Alliances were created with partners (no longer considered 'contractors' or 'suppliers') who brought their expertise to bear on the problems of development.

The traditional approach to learning, because it was institutionalized, had to be discredited. The example in Box 1 gives a good example of how the symbols which validate this learning—the studies, the role of the manager in approving the studies, the post-project audit—can be discredited by an effective leader. Rhetoric can be a very effective strategic tool.⁸

Behaviours must change to support this new approach to learning. The discovery of Foinaven, for example, the first commercial discovery in the Atlantic, and an asset likely to be worth in excess of US\$2 billion, would probably not have occurred without behavioural change. The exchange of technical information on new seismic techniques between BP's Gulf of Mexico and Atlantic assets was encouraged by international exchanges, and the willingness of individuals to experiment with alien

Institutionalizing this new approach to learning

TABLE 2. The new approach to learning

CHARACTERISTICS	OUTCOMES
• Recognizing that uncertainty is avoidable • Emphasizing key success factors • Designing 'uncertainty intervention' strategies • Action learning • Cross-functional structures and lateral linkages • Recognition of information leakage • Accessing best knowledge and skills through alliances and external benchmarking	• Lower appraisal and development costs • Flexible routing in appraisal and development • Parallel approach to appraisal, development and production • Much reduced time to first oil (less than half) • Increased present value of projects

techniques, only happened because space for individual initiative had been created in the organization.⁹

New behaviours were validated by new symbols and meanings. The action orientation of learning encouraged at BP was captured in the name of the change consultants employed—JMW Consultants ('Just Might Work'), and in the 'JDI' (Just Do It) awards made to offshore operational personnel. In the same way, validation of learning from external sources was recognized in the admission that parallel exploration, appraisal and development is a concept leveraged from Toyota's parallel engineering approach to new automobile model development. Such approaches help break down apparent industry paradoxes, and provide new innovative solutions to development.⁹

In all three case studies, the role of the leader in enabling or encouraging new learning behaviours appears to have been critical. BP's asset manager, for example, described his job as designing the 'organizational technology' to allow innovative learning.

The Pace of Learning

Stalk¹⁰ asserted that 'time' was the next source of competitive advantage. The three case studies reported here demonstrate a growing understanding of *the significance of time* to the successful and profitable outcome of the projects.

The parallel engineering approach followed at Foinaven (see Fig. 3) successfully brought first oil forward by over five years, and this approach was being followed in the development of the Schehalion prospect. BP management estimated the present value benefit of bringing revenue flows forward in this way, as applied to the UKCS portfolio of prospects, as being in excess of £200 million. This change of approach was matched by a switch in the measure for describing the significance of a

prospect: traditionally, Foinaven would have been measured by millions of barrels of recoverable crude oil, or crude oil equivalent; the Foinaven project saw the description of scale measures change to estimated present values.

BP and its 50% Atlantic joint venture partner, Shell, have also seen the development of existing prospects at Foinaven and Schehalion as weapons in the market to develop the future of this frontier province. Thus, extensive wide area 3-D seismic data were acquired, which, together with the experience of developing the first commercial field at

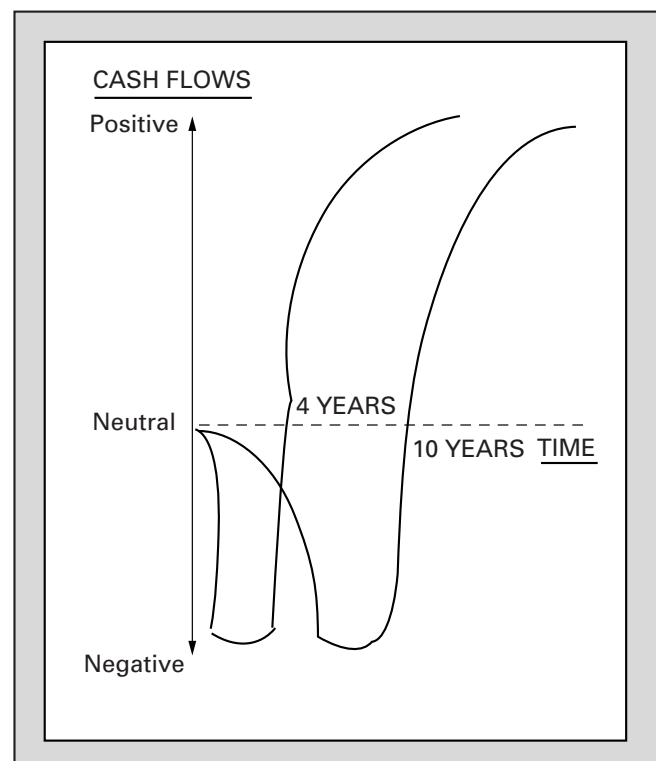


FIGURE 3. The typical effect of parallel development engineering on project cash flows.

Foinaven appeared to position them well in the bidding for future licence prospects in 1994. Commentators at that time suggested that the established credibility of the BP–Shell joint venture encouraged the UK government to award a majority of further exploration prospects to the joint venture.

The Alliance partners at the development project on the Andrew field were likewise concerned with time as an important dimension of value. This surfaced in two obvious examples from their innovative development.

- ❑ The partners ascribe beating the development timetable to first oil by nearly six months to the *more effective ways of working they developed as an Alliance*. Some of this can be attributed to the efficiency of networked flows of information and knowledge in an Alliance structure.¹¹
- ❑ There was considerable emphasis on re-engineering the phasing of the development to minimize the installation time spent offshore. For example, hooking up the offshore production units was completed in a matter of minutes rather than the previous best practice achievements of ten to twenty hours. *Minimizing offshore installation time* was estimated by the Andrew Alliance to have saved £20 million from the development cost of the field.

The focus on development time and the emphasis on learning in these companies led to an awareness of the importance of the pace of learning. The dynamic aspect of learning captured by the Boston Consulting Group⁴ in their studies of the experience curve remains the best known example of this effect, although the link between the rate of learning and cumulative output volumes has been questioned.¹² Building on the ideas concerning core competence of Prahalad and Hamel⁶, Leavy¹³ describes *learning as “in some respects analogous to money ... its impact depends not only on the stock of knowledge but also on the velocity of its circulation”*.

In these three oil field development projects, the pace of learning was beneficial in three ways:

- ❑ The pace of learning impacted the date of first oil. In the case of Foinaven, this meant that learning new 3-D seismic techniques that could allow an accurate delineation of sandstone reservoirs eliminated the need for a time-consuming and costly programme of drilling appraisal wells.
- ❑ Such learning is transferable to other projects—interpretation techniques on this 3-D seismic were transferred to Foinaven by a manager seconded from the Gulf of Mexico, and the same appraisal techniques will reduce the development cost of the Schehalion field.

- ❑ Learning as a capability is enhanced through its use. On these projects, learning capability includes the ability to encounter new and possibly unique issues or success factors, and to design intervention strategies to deal with them. The pace of learning is therefore significant to how quickly a company such as BP can enhance its distinctive learning capabilities for future exploration and development projects, and build competitive advantage.⁹ As with BCG’s experience curve, the speed at which learning capability is acquired determines the likelihood that that capability might remain a source of advantage (see Box 2).

Box 2

Pace of learning and improved performance

“It’s not just the learning—it’s not just being able to sit back after the action and say I learnt the following, and I will write a recipe book so that someone else doesn’t make the same mistake or that someone else can do it. The message coming out of Foinaven is its the pace at which you learn which matters. It is pointless to have a book at the end of Foinaven with all the lessons written in it—the lessons are learnt daily and they’re put into action immediately. So it’s the pace of learning on Foinaven, which enhances our company’s performance and the next projects we do West of Shetland....

It’s fundamental to our success as a business that we create the industry in a way, which will provide more profitable outcomes. The key to this is continual ongoing education and support of people so they can take on bigger risks and understand the consequences of taking these risks.”

Colin MacLean, BP Asset Manager

Structuring Learning to Maximize Returns

The three case studies demonstrate differences in the way learning occurs at different stages of the upstream business which, in their own ways, help to maximize the overall potential returns to oil companies.

Legally, superior returns and added value for upstream oil companies seem to derive from the exclusive rights¹⁴ to exploit oil and gas properties. Such strategic assets are, however, depreciating physical resources, and competitive advantage over the long run is more likely to be embedded in the

capabilities that enhance an oil company's ability to acquire new properties in competition with other bidders. In particular, oil companies have increasingly realized that their most likely sources of competitive advantage lie in their acquisition and manipulation of private information and knowledge concerning prospective oil and gas properties. The competitive advantage from such knowledge asserts itself during the bidding stages for new exploration prospects and, later, as the results of exploration are digested, in trading properties with other companies or bidding for related properties.

Secrecy surrounding such data and knowledge is paramount. Learning in this area of activity is therefore highly competitive, and, as far as is possible, secured within the organization. It was, for example, notable in the study that staff explaining some of the seismic interpretation, which enabled the identification, and speedy appraisal of Foinaven provided broad outlines only of the techniques.

In the areas of developing discovered properties and producing oil from them, however, competitive advantage is less likely to be capable of being secured. Oil companies have traditionally controlled such development projects, but relying heavily on contractors and suppliers. Leakage of private information during these activities is always a risk, and dispersal of any skills-based advantage is highly likely, therefore, at this stage. BP and its alliance partners have responded by recognizing that maximizing returns in these activities does not result from the pursuit of competitive advantage. Rather, it is increasingly accepted that maximizing returns to the industry in these activities is the best route to improved returns for the company. This is considered to be a necessary response, also, in an sector where the profitability of assets declined precipitously after the collapse of the crude oil price from \$26 per barrel, to an average \$15 to \$18 per barrel in the period since 1986.

There has, therefore, been a restructuring of development activities to recognize the benefits of a collaborative approach. Techniques for partner alignment have radically restructured the reward and risk profile of fields where alliancing is in place. The Andrew field in this case study stands as the pioneer of this approach in the UK Continental Shelf. *The strategic learning benefits of alliancing on Andrew* included: the widening of the expert opinion that could be drawn on in developing the field; a further increase in the pace of learning as network learning took the place of hierarchical relationships between client oil companies, prime contractors and sub-contractors; and the often easier access to technology that previously had been guarded as in some way proprietary. These learning benefits helped to deliver the Andrew project six months ahead of schedule and £80 million (or 21%)

under an already demanding target budget.¹⁵ The collaborative arrangements meant that BP's Alliance partners shared 56% of this saving.

Alliancing as a capability in its own right has, of course, become a focus of competitive activity, with suppliers and oil companies promoting themselves as preferred allies on the basis of the experience they have gained on projects like the Andrew Alliance. This has led to a shift in the focus of development projects from the opportunities of technology, to those of organizational and inter-organizational process:

BP also acknowledges that any technological lead ... is likely to be short-lived in an industry where partnerships and the sharing of information is the rule ... "Anybody can copy a formulation of your technology," says Mr Simon (BP's former CEO), "but your competitors cannot ape your process".¹⁶

Despite this focus on the unique structures of alliancing and the pace of collaborative learning as distinctive capabilities in their own right, the distinction between learning in exploration activities and the collaborative open approach to learning in the development activity is relatively clear. In the former instance, learning represents a potential source of advantage, in the latter a means to make the industry generally more attractive. Companies like BP can benefit simultaneously from these two routes to better returns.

Conclusions

The upsurge in interest in organizational learning is timely given the current focus in the resource-based view of strategy on capabilities as the primary source of sustainable competitive advantage. Leavy,¹³ however, notes that some of the writing on organizational learning and knowledge is couched in the language of 'a humanistic mission', citing both Senge¹ and Nonaka¹⁷ as examples where the learning organization is said to produce economic and 'more spiritual' benefits. Pedler et al.² draw heavily on Lessem¹⁸ to propose a four-stage organizational development model culminating in a 'developmental', 'integrated' organization—the 'Learning Company'. The danger in such missions, however, is the tendency to see all learning as of necessity a good thing. *Learning can be as much a core rigidity as a core competence.*

This article has been concerned, however, to draw lessons from the three case studies in identifying the strategic significance of learning in the offshore oil industry.

The cases identify the importance of institutionalizing appropriate approaches to learning. Traditional institutional approaches, which do not take proper account of the uncertainty, risk and

value profiles of oil field development projects, were identified and changed.

All the cases demonstrate the significance of pace of learning as a potential source of advantage for competition in time. Pace of learning was seen as significant in all three cases, both for the dramatic improvement in project value which management achieved on the projects themselves, and also for the faster development of strategic learning capabilities which can prove to be an ongoing source of advantage.

The strategies for learning developed for the different projects demonstrated a sensitivity to the

best way of improving returns for individual companies. In the key exploration areas information and knowledge can be a proprietary source of advantage, therefore secured organizational learning can also be valuable. In the appraisal and development activities, however, a competitive advantage seems more difficult to establish and to sustain, and an open approach, which embraces alliancing, may do more to improve company returns by improving industry returns. This dual competitive and collaborative approach to maximizing company returns has obvious application to most industries.

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